CONSENT TO PARTICIPATE IN RESEARCH

Title of Study: Cognitive Neuroscience Studies of Action and Cognition
Consent to Participate in a Research Study (for Patients with Neurological Disorders)

Key Information

- You are being invited to participate in a research study. Participation in research is completely voluntary.
- The purpose of the study is to examine how people acquire and produce certain "motor skill" behaviors and how visual and auditory information are used to guide actions.
- The study will take a total of one to two hours and you will be asked to read the consent form and answer questions about it.
- Risks and/or discomforts may include the rare risk of frustration with the task.
- There is no direct benefit to you. We expect this research to lead to a better understanding of their problems and hope that this knowledge will eventually improve treatment for patients with neurological problems.

Introduction

My name is Professor Richard Ivry. I am a faculty member in the Psychology Department at the University of California, Berkeley. I invite you to take part in a research study.

Purpose

The research deals with how people acquire and produce certain "motor skill" behaviors and how visual and auditory information are used to guide actions, as well as how the specialization of different parts of your brain can affect your performance. The study may involve carrying out different tasks involving motor skills. We are interested in how injury to the brain affects performance on these tasks. You are being asked to take part in this study because you have a disorder or injury to your brain.

Procedures

- If you agree to participate, this is what will happen: You will be seated in front of a computer monitor. The computer will present visual displays or auditory signals that are used to indicate the types of response you are supposed to produce. The manner of your response will vary depending on the experiment you have been recruited for. You may be asked to:
    a) Press keys on a keyboard, using the fingers of your hand.
    b) Make arm movements to target locations on a table surface. We will place small markers on your index finger to track the position of your arm.
    c) Make arm movements which are recorded using depth cameras.
    d) Make eye movements which are recorded using an infrared eye tracking camera.
    e) Make perceptual judgments of your arm of hand location while vibration is applied to the tendon of that limb. This will be done by placing a muscle vibrator (massaging

device normally used for physical therapy) over muscles in your upper or lower arm and vibrating them for 1/2 to 3 minutes while you keep your eyes closed.

f) In linguistic tasks you will produce a spoken response to stimuli which will be recorded.

g) For some of these tasks, we may also use EMG (electromyographic) instruments to record muscle activity in your arm or hand. This involves lightly swabbing your skin with fine-grain sandpaper and alcohol, and then pasting small electrodes to the surface of your skin. (Electrodes are small, thin metal discs, about the size of a dime, that detect tiny charges resulting from muscle activity).

h) In studies that measure brain waves with electroencephalography (EEG), we will fit a cap with electrodes on your head. The electrodes will rest on your scalp to measure your brain activity. Gel will be inserted into each electrode to improve its measuring sensitivity. This preliminary set-up can take between 30-40 minutes. Once testing is over, you will be given a place to clean and fix your hair, as the cap and gel can leave a paste on your hair where the electrodes were placed. The gel will wash out completely without harming your hair or scalp.

i) If you agree, we may videotape or audio record you undergoing some of these procedures.

j) You may be administered different neuropsychological batteries. These include established and well-validated neuropsychological tests that assess cognitive functions like attention, reasoning, memory and motor abilities.

Study time: The test session will take approximately one to two hours. You will be given breaks every 10 minutes or so, or more often if you request them.

Study location: These studies will be conducted in my laboratory at UC Berkeley, collaborative sites at laboratories at UC San Francisco, VA Medical Center in Martinez, at the National Ataxia Foundation Annual Conference, or at your home. All studies will be conducted under the supervision of Prof. Ivry.

Benefits

It is important that you understand that there is no direct benefit to you anticipated from your participation in this study. Healthy people, such as yourself, provide us with the standard performance on these tasks. The study is not related to your medical treatment. However, we hope to obtain knowledge from this research about how the brain works which could prove useful in the future for treating patients with neurological problems similar to yours.

Risks/Discomforts

It is possible but unlikely that doing any of the tasks or using the devices described above could cause you discomfort. The visual and auditory stimuli are presented at a comfortable lighting and sound level and the movements require a minimal amount of effort. These procedures have been used extensively in previous research, and we have taken precautions to prevent any discomforts or malfunctions of the devices. There is a slight chance you may feel frustrated or uncomfortable if some of the tasks are difficult for you to do, but you are free to discontinue your participation in the study if you choose. Please let us know if you feel fatigued or uncomfortable at any time during the experiment.

- Behavioral Task: There is a slight chance you may feel frustrated or uncomfortable if some of the tasks are difficult for you to do, but you are free to discontinue your participation in the study if you choose. Please let us know if you feel fatigued or uncomfortable at any time during the experiment.

- EMG: Many people have been studied using EMG instruments without reported harm. There is a slight chance of skin irritation or dryness due to the sandpaper and alcohol pretreatment. If this occurs, we will provide you with an ointment to reduce skin irritation.
- EEG: Many people have been studied using EEG techniques without reported harm. However, there is a small risk of electric shock associated with the electrodes. We have taken precautions to minimize this risk by using battery operated amplifiers that are completely isolated from any electrical outlet. In addition, both you and the entire EEG system will be grounded before the experiment to ensure minimal risk. There is a small risk of disease transmission through the electrode cap. In order to minimize this risk strict standardized procedures are used to disinfect the cap after each use. After each testing session, the cap and electrodes are saturated with an anti-viral, anti-bacterial, antifungal solution for 10 minutes. The cap is then washed in warm soapy water to destroy possible germs. All individuals that sterilize the cap and electrodes wear latex gloves during the whole process. It is also possible that the EEG cap can cause discomfort during the testing session. If this occurs, please let us know so that we can immediately adjust the cap and add padding to minimize this discomfort.

Confidentiality

Your information will be handled as confidentially as possible. If results of this study are published or presented, individual names and other personally identifiable information will not be used, unless you have specified your permission for this in the attached Media Records Release consent form.

To minimize the risks to confidentiality, your results will be coded. Study data and the key to the code for subject identities will be kept in separate, locked files, and only the experimenter, principal investigator, and lab coordinator will have access to these files. Your data will be encrypted and securely stored indefinitely on password-protected computers. In order to award academic credit, the RPP will initially retain your student ID number but this will be deleted at the end of the semester.

Your personal information may be released if required by law. Authorized representatives from the following organizations may review your research data for purposes such as monitoring or managing the conduct of this study: National Institutes of Health, University of California, Food and Drug Administration (FDA) and other government agencies involved in keeping research safe for people.

Retaining research records: When the research is completed, I will save the data for possible use in future research done by myself or others. I will retain this study information indefinitely after the study is over. The same measures described above will be taken to protect confidentiality of this study data.

Identifiers might be removed from the identifiable private information or identifiable biospecimens. After such removal, the information or biospecimens could be used for future research studies or distributed to other investigators for future research studies without additional informed consent from the subject or the legally authorized representative.

This research is covered by a Certificate of Confidentiality from the National Institutes of Health. This means that the researchers cannot release or use information, documents, or samples that may identify you in any action or suit unless you say it is okay. They also cannot provide them as evidence unless you have agreed. This protection includes federal, state, or local civil, criminal, administrative, legislative, or other proceedings. An example would be a court subpoena.

There are some important things that you need to know. The Certificate DOES NOT stop reporting that federal, state or local laws require. Some examples are laws that require reporting of child or elder abuse, some communicable diseases, and threats to harm yourself or others. The Certificate CANNOT BE USED to stop a sponsoring United States federal or state government agency from checking records or evaluating programs. The Certificate DOES NOT stop disclosures required by the federal Food and Drug

Administration (FDA). The Certificate also DOES NOT prevent your information from being used for other research if allowed by federal regulations.

Researchers may release information about you when you say it is okay. For example, you may give them permission to release information to insurers, medical providers or any other persons not connected with the research. The Certificate of Confidentiality does not stop you from willingly releasing information about your involvement in this research. It also does not prevent you from having access to your own information.

Compensation/Payment

You will be paid at the rate of $20 per hour plus reimbursed for any travel expenses involved in your study participation. A check will be sent to your address within two weeks of completion of the study.

Rights

Participation in research is completely voluntary. You have the right to decline to participate or to withdraw at any point in this study without penalty or loss of benefits to which you are otherwise entitled.

Questions

If you have any questions about this research, please call me, Professor Richard Ivry, at 510-642-7146 (office) or 510-642-0135 (lab), or email me at ivry@berkeley.edu.

If you have any questions about your rights or treatment as a participant in this research project, please contact the University of California at Berkeley's Committee for Protection of Human Subjects at 510-642-7461 or subjects@berkeley.edu.

CONSENT

You will be given a copy of this consent form and of the Medical Research Subject's Bill of Rights to keep.

If you wish to participate in this study, please sign and date below.

_____
Participant's Name *(please print)*

_____ _____
Participant's Signature Date

_____ _____
Person Obtaining Consent Date